

Army pool sensitivity — complete run matrix checklist

Purpose: Compare how **pool membership rules**, excluding **zero–talent-board peers**, and **peer context metric on plots** change **sorting index** (panel 5, H_sort) and **promotion-rate porch** (panel 9, HERO).

Where you work: AWS 520 root — the folder named `Network_1P_shell/` that contains:

- `big_dfs/`
- `520_pipeline_cox_working.ipynb`
- `pipeline_config.py`
- `talent/re_entry/` (plot scripts and manifest)
- `scripts/backup_rename_suffix.sh`

What you compare for Alex (assortativity priority):

- Panel 5 — read **H_sort** in the overlap plot title.
- Panel 9 — read **HERO** porch shape (promotion rate vs peer-context bins).
- Record values in the **results table** at the end of this document.

Vocabulary (read once)

Plain English	Setting or column name
Legacy pool grain	<code>POOL_GROUPING_MODE = 'legacy'</code> — peers share the same HRC snapshot date and the same senior-rater identifier (<code>snr_rater_bwd</code>).
Eval-thru pool grain	<code>POOL_GROUPING_MODE = 'active_at_eval_thru'</code> — peers are senior-rater officers whose OER evaluation window covers this row’s OER completion date (<code>eval_thru_dt_bwd</code>).
Keep zero–TB peers in pool	<code>POOL_EXCLUDE_PEER_TB_ZERO = False</code>
Drop zero–TB peers from pool mean and size	<code>POOL_EXCLUDE_PEER_TB_ZERO = True</code> — removes peers with <code>tb_ratio == 0</code> from pool statistics; does not remove the rated officer from the cohort.
Straight leave-one-out pond level (T_j)	Feather column <code>pool_tb_ratio_mean_snr_fwd</code> — mean senior-rater pool talent, leave-one-out.
Minus-mean peer context	Feather column <code>pool_minus_mean_snr_fwd</code> — own talent minus pool mean.
Z-scored minus-mean (HERO default for minus-mean runs)	Feather column <code>z_pool_minus_mean_snr_fwd</code>
Live feather file plots read	<code>big_dfs/df_pipeline_11_cox_analysis.feather</code>
Data story mosaic output	<code>talent/re_entry/output/data_story/ARMY_DATA_STORY_run1_3x3.png</code>

Plot tag `_run1` in filenames: All five runs write PNGs with suffix `_run1` in the filename. That is a **plot tag**, not “Run 1 only.” Each sensitivity run **overwrites** the top-level PNGs until you **backup** into a subfolder.

Five runs — overview

Run label	Pool grain	Zero–TB peers	Peer metric on plots	Re-run notebook?
Run 1	legacy	keep	straight leave-one-out pond level	Yes
Run 2	eval-thru	keep	straight leave-one-out pond level	Yes
Run 3	eval-thru	exclude from pool	straight leave-one-out pond level	Yes
Run 4 (optional)	legacy (from archived feather)	keep	minus-mean	No — plots only
Run 5 (optional)	eval-thru (from archived feather)	keep	minus-mean	No — plots only

Feather archives you must save after notebook runs:

Archive file	Create after
<code>big_dfs/df_pipeline_11_run1_legacy.feather</code>	Run 1 notebook
<code>big_dfs/df_pipeline_11_run2_thru.feather</code>	Run 2 notebook
<code>big_dfs/df_pipeline_11_run3_thru_nozero.feather</code>	Run 3 notebook

Plot backup (`backup_rename_suffix.sh`) **does not** copy feathers. You must run the `cp` commands in each run’s save step.

Shared reference — plot script lines

These files live under `talent/re_entry/` on AWS.

Straight leave-one-out pond level (Runs 1, 2, 3)

File	Line	Must read
army_basic_plots.py	42	COL_L00 = "pool_tb_ratio_mean_snr_fwd"
army_hero_slide_plot.py	44	plot_var = "pool_tb_ratio_mean_snr_fwd"
army_act2_probes.py	55	COL_L00_Z = "z_pool_tb_ratio_mean_snr_fwd" (column usually missing → Act II z-scores COL_L00 from basic plots)

HERO command-line flag (Runs 1–3):

```
python talent/re_entry/army_hero_slide_plot.py --plot-var pool_tb_ratio_mean_snr_fwd
```

Manifest panel 9 path (Runs 1–3):

```
talent/re_entry/output/hero/ARMY_HERO_ew8_pool_tb_ratio_mean_snr_fwd_run1.png
```

Do not use `--plot-var pool_tb_ratio_mean_snr_fwd_run1.png` — that is a filename, not a column name.
Do not use `z_pool_tb_ratio_mean_snr_fwd` for Runs 1–3 unless you add that column via a separate Cell 10.5 config path.

Minus-mean peer context (Runs 4, 5)

File	Line	Must read
army_basic_plots.py	42	COL_L00 = "pool_minus_mean_snr_fwd"
army_hero_slide_plot.py	44	plot_var = "z_pool_minus_mean_snr_fwd"
army_act2_probes.py	55	COL_L00_Z = "z_pool_minus_mean_snr_fwd"

HERO command-line flag (Runs 4, 5):

```
python talent/re_entry/army_hero_slide_plot.py --plot-var z_pool_minus_mean_snr_fwd
```

Manifest panel 9 path (Runs 4, 5):

```
talent/re_entry/output/hero/ARMY_HERO_ew8_z_pool_minus_mean_snr_fwd_run1.png
```

Plot pipeline commands (every run that builds plots)

Run from `Network_1P_shell/`:

```
python talent/re_entry/army_basic_plots.py --all
python talent/re_entry/army_act2_probes.py --plot all_probes
python talent/re_entry/army_hero_slide_plot.py --plot-var <COLUMN_NAME>
python talent/re_entry/build_army_data_story.py
```

Use `python`, not `bash`, for `build_army_data_story.py`.
Success lines:

- Wrote talent/re_entry/output/hero/ARMY_HERO_ew8...
- Wrote talent/re_entry/output/data_story/ARMY_DATA_STORY_run1_3x3.png

Manifest file

Path: `talent/re_entry/manifests/army_run1_3x3_manifest.json`
Edit panel 9 `"path"` to match the HERO PNG filename **exactly** before running `build_army_data_story.py`.
Panels 2–8 paths in the manifest stay the same; scripts overwrite top-level PNGs.

Notebook cell order (every run that re-runs the notebook)

- Open `520_pipeline_cox_working.ipynb`. Run in order:
- Cell 0 — reloads `pipeline_config.py`
 - Cell 5 — pool means (watch log for `POOL_GROUPING_MODE`)
 - Cell 6 — or if Cell 6 fails, run in terminal:
`cp big_dfs/df_pipeline_05_pool_means.feather big_dfs/df_pipeline_06_pool_ranks.feather`
 - Cell 7
 - Cell 8
 - Cell 9
 - Cell 10
 - Cell 10.5
 - Cell 11 — writes `big_dfs/df_pipeline_11_cox_analysis.feather`

Feather column check (after Cell 11)

```
python3 -c "import pandas as pd; df=pd.read_feather('big_dfs/df_pipeline_11_cox_analysis.feather');
print('pool_tb_ratio_mean_snr_fwd', 'YES' if 'pool_tb_ratio_mean_snr_fwd' in df.columns else 'MISSING')"
```

For Runs 4 and 5 also check:

```
python3 -c "import pandas as pd; df=pd.read_feather('big_dfs/df_pipeline_11_cox_analysis.feather');
print('z_pool_minus_mean_snr_fwd', 'YES' if 'z_pool_minus_mean_snr_fwd' in df.columns else 'MISSING')"
```

RUN 1 • Legacy pool grain • straight leave-one-out • keep zero–TB peers

What this run isolates: Baseline legacy pool definition with pond-level peer context on plots.

Run 1 — Step 1 • Open terminal on AWS

```
cd /path/to/Network_1P_shell
conda activate TALNET39
date
ls -lh big_dfs/df_pipeline_11_cox_analysis.feather
```

The last line is optional; it shows whether an old feather exists before you rebuild.

Run 1 — Step 2 • Edit pipeline_config.py

Open pipeline_config.py at 520 root. Set these lines exactly:

```
POOL_GROUPING_MODE = 'legacy'
POOL_EXCLUDE_PEER_TB_ZERO = False
CELL5_POOL_MEANS = True
CELL6_POOL_RANKS = True
```

Save the file.

Also confirm these lines exist (needed by Cell 5; add if missing):

```
POOL_ANCHOR_COL = 'eval_thru_dt_bwd'
POOL_EVAL_STRT_COL = 'eval_strt_dt_bwd'
POOL_EVAL_THRU_COL = 'eval_thru_dt_bwd'
```

Run 1 — Step 3 • Edit plot scripts (straight leave-one-out)

File: talent/re_entry/army_basic_plots.py — line 42:

```
COL_L00 = "pool_tb_ratio_mean_snr_fwd"
```

File: talent/re_entry/army_hero_slide_plot.py — line 44:

```
plot_var = "pool_tb_ratio_mean_snr_fwd"
```

File: talent/re_entry/army_act2_probes.py — line 55 (optional but recommended for consistent panels 7–8):

```
COL_L00_Z = "z_pool_tb_ratio_mean_snr_fwd"
```

Save all edited files.

Run 1 — Step 4 • Run notebook cells

Open 520_pipeline_cox_working.ipynb .

Run Cell 0, then Cell 5, Cell 6 (or the cp fallback), Cell 7, Cell 8, Cell 9, Cell 10, Cell 10.5, Cell 11.

In the Cell 5 log, confirm:

```
POOL_GROUPING_MODE='legacy'
```

Run 1 — Step 5 · Check feather columns

```
python3 -c "import pandas as pd; df=pd.read_feather('big_dfs/df_pipeline_11_cox_analysis.feather');
print('pool_tb_ratio_mean_snr_fwd', 'YES' if 'pool_tb_ratio_mean_snr_fwd' in df.columns else 'MISSING')"
```

Expect: **YES**.

Run 1 — Step 6 · Edit manifest panel 9

Open `talent/re_entry/manifests/army_run1_3x3_manifest.json`.

Set panel 9 `"path"` to:

```
talent/re_entry/output/hero/ARMY_HERO_ew8_pool_tb_ratio_mean_snr_fwd_run1.png
```

Save the manifest.

Run 1 — Step 7 · Run plot scripts and mosaic

```
python talent/re_entry/army_basic_plots.py --all
python talent/re_entry/army_act2_probes.py --plot all_probes
python talent/re_entry/army_hero_slide_plot.py --plot-var pool_tb_ratio_mean_snr_fwd
python talent/re_entry/build_army_data_story.py
```

Run 1 — Step 8 · Verify Run 1

```
grep POOL_GROUPING_MODE pipeline_config.py
ls -l talent/re_entry/output/hero/ARMY_HERO_ew8_pool_tb_ratio_mean_snr_fwd_run1.png
ls -l talent/re_entry/output/data_story/ARMY_DATA_STORY_run1_3x3.png
```

In the basic plots log, find:

```
Overlap pool grain: POOL_GROUPING_MODE='legacy'
```

Open panel 5 — note **H_sort**. Open panel 9 — note porch shape.

Compare mosaic to standalone PNGs at **top level** under `talent/re_entry/output/` (not backup subfolders).

Run 1 — Step 9 · Archive feather and plots before Run 2

Save feather (required for Run 4 later):

```
cp big_dfs/df_pipeline_11_cox_analysis.feather big_dfs/df_pipeline_11_run1_legacy.feather
```

Save plots:

```
bash scripts/backup_rename_suffix.sh _run1_legacy --all-output
```

RUN 2 · Eval-thru pool grain · straight leave-one-out · keep zero—TB peers

What this run isolates: Same plot peer metric as Run 1, but **eval-thru** pool grain — tests **assortativity** / **H_sort** effect of pool definition.

Run 2 — Step 1 · Open terminal on AWS

```
cd /path/to/Network_1P_shell
conda activate TALNET39
date
```

Run 2 — Step 2 · Edit `pipeline_config.py`

```
POOL_GROUPING_MODE = 'active_at_eval_thru'
POOL_EXCLUDE_PEER_TB_ZERO = False
CELL5_POOL_MEANS = True
CELL6_POOL_RANKS = True
```

Save the file.

Run 2 — Step 3 · Edit plot scripts (straight leave-one-out)

File: `talent/re_entry/army_basic_plots.py` — line 42:

```
COL_L00 = "pool_tb_ratio_mean_snr_fwd"
```

File: `talent/re_entry/army_hero_slide_plot.py` — line 44:

```
plot_var = "pool_tb_ratio_mean_snr_fwd"
```

File: `talent/re_entry/army_act2_probes.py` — line 55:

```
COL_L00_Z = "z_pool_tb_ratio_mean_snr_fwd"
```

Save all edited files.

Run 2 — Step 4 · Run notebook cells

Open `520_pipeline_cox_working.ipynb`.

Run Cell 0, Cell 5, Cell 6 (or `cp` fallback), Cell 7, Cell 8, Cell 9, Cell 10, Cell 10.5, Cell 11.

In the Cell 5 log, confirm:

```
POOL_GROUPING_MODE='active_at_eval_thru'
```

Run 2 — Step 5 · Check feather columns

```
python3 -c "import pandas as pd; df=pd.read_feather('big_dfs/df_pipeline_11_cox_analysis.feather');
print('pool_tb_ratio_mean_snr_fwd', 'YES' if 'pool_tb_ratio_mean_snr_fwd' in df.columns else 'MISSING')"
```

Expect: `YES`.

Run 2 — Step 6 · Edit manifest panel 9

Open `talent/re_entry/manifests/army_run1_3x3_manifest.json`.

Set panel 9 `"path"` to:

```
talent/re_entry/output/hero/ARMY_HERO_ew8_pool_tb_ratio_mean_snr_fwd_run1.png
```

Save the manifest.

Run 2 — Step 7 · Run plot scripts and mosaic

```
python talent/re_entry/army_basic_plots.py --all
python talent/re_entry/army_act2_probes.py --plot all_probes
python talent/re_entry/army_hero_slide_plot.py --plot-var pool_tb_ratio_mean_snr_fwd
python talent/re_entry/build_army_data_story.py
```

Run 2 — Step 8 · Verify Run 2

```
grep POOL_GROUPING_MODE pipeline_config.py
ls -l talent/re_entry/output/hero/ARMY_HERO_ew8_pool_tb_ratio_mean_snr_fwd_run1.png
ls -l talent/re_entry/output/data_story/ARMY_DATA_STORY_run1_3x3.png
```

In the basic plots log, find:

```
Overlap pool grain: POOL_GROUPING_MODE='active_at_eval_thru'
```

Compare panel 5 `H_sort` and panel 9 porch to Run 1 backup in `talent/re_entry/output/*/run1_legacy/`.

Run 2 — Step 9 · Archive feather and plots before Run 3

Save feather (required for Run 5 later):

```
cp big_dfs/df_pipeline_11_cox_analysis.feather big_dfs/df_pipeline_11_run2_thru.feather
```

Save plots:

```
bash scripts/backup_rename_suffix.sh _run2_thru --all-output
```

RUN 3 · Eval-thru pool grain · straight leave-one-out · exclude zero–TB peers

What this run isolates: Same eval-thru grain as Run 2, but peers with **talent board ratio equal to zero** are dropped from pool mean and pool size.

Run 3 — Step 1 · Open terminal on AWS

```
cd /path/to/Network_1P_shell
conda activate TALNET39
date
```

Run 3 — Step 2 · Edit `pipeline_config.py`

```
POOL_GROUPING_MODE = 'active_at_eval_thru'
POOL_EXCLUDE_PEER_TB_ZERO = True
CELL5_POOL_MEANS = True
CELL6_POOL_RANKS = True
```

Save the file.

Run 3 — Step 3 · Edit plot scripts (straight leave-one-out)

File: `talent/re_entry/army_basic_plots.py` — line 42:

```
COL_L00 = "pool_tb_ratio_mean_snr_fwd"
```

File: `talent/re_entry/army_hero_slide_plot.py` — line 44:

```
plot_var = "pool_tb_ratio_mean_snr_fwd"
```

File: `talent/re_entry/army_act2_probes.py` — line 55:

```
COL_L00_Z = "z_pool_tb_ratio_mean_snr_fwd"
```

Save all edited files.

Run 3 — Step 4 · Run notebook cells

Open `520_pipeline_cox_working.ipynb`.

Run Cell 0, Cell 5, Cell 6 (or `cp` fallback), Cell 7, Cell 8, Cell 9, Cell 10, Cell 10.5, Cell 11.

In the Cell 5 log, confirm:

```
POOL_GROUPING_MODE='active_at_eval_thru'
```

and, if printed:

```
POOL_EXCLUDE_PEER_TB_ZERO=True
```

Run 3 — Step 5 · Check feather columns

```
python3 -c "import pandas as pd; df=pd.read_feather('big_dfs/df_pipeline_11_cox_analysis.feather');
print('pool_tb_ratio_mean_snr_fwd', 'YES' if 'pool_tb_ratio_mean_snr_fwd' in df.columns else 'MISSING')"
```

Expect: `YES`.

Optional pool-size diagnostic:

```
python -u talent/re_entry/army_pool_size_probe.py
```

Run 3 — Step 6 · Edit manifest panel 9

Open `talent/re_entry/manifests/army_run1_3x3_manifest.json`.

Set panel 9 "path" to:

```
talent/re_entry/output/hero/ARMY_HERO_ew8_pool_tb_ratio_mean_snr_fwd_run1.png
```

Save the manifest.

Run 3 — Step 7 · Run plot scripts and mosaic

```
python talent/re_entry/army_basic_plots.py --all
python talent/re_entry/army_act2_probes.py --plot all_probes
python talent/re_entry/army_hero_slide_plot.py --plot-var pool_tb_ratio_mean_snr_fwd
python talent/re_entry/build_army_data_story.py
```

Run 3 — Step 8 · Verify Run 3

Panel 5 title should include `exclude_tb_ratio=0 peers`.

Compare panel 5 H_sort and panel 9 porch to Run 2 backup in `talent/re_entry/output/*/run2_thru/`.

Panel 2 left histogram may still show a spike at own talent equal to zero — that is **own performance**, not the pool-exclusion toggle.

Run 3 — Step 9 · Archive feather and plots

Save feather:

```
cp big_dfs/df_pipeline_11_cox_analysis.feather big_dfs/df_pipeline_11_run3_thru_nozero.feather
```

Save plots:

```
bash scripts/backup_rename_suffix.sh _run3_thru_nozero --all-output
```

RUN 4 · Legacy pool grain · minus-mean plots · plot-only (no notebook)

What this run isolates: Minus-mean peer context on plots, using the **same feather as Run 1** (legacy grain). **Do not re-run the notebook** unless the Run 1 archive is missing.

Run 4 — Step 1 · Open terminal on AWS

```
cd /path/to/Network_1P_shell
conda activate TALNET39
date
ls -lh big_dfs/df_pipeline_11_run1_legacy.feather
```

If the archive file **does not exist**, stop and complete **Run 1** through **Run 1 Step 9** first.

Run 4 — Step 2 · Restore Run 1 feather into the live slot

Plot scripts always read `big_dfs/df_pipeline_11_cox_analysis.feather`. After Run 3, that file holds Run 3 data. Copy the Run 1 archive back:

```
cp big_dfs/df_pipeline_11_run1_legacy.feather big_dfs/df_pipeline_11_cox_analysis.feather
```

Direction: `run1_legacy` → `cox_analysis` (restore, not save).

Run 4 — Step 3 · `pipeline_config.py`

No edit required for plot-only Run 4 — pool columns are already in the restored feather.

Run 4 — Step 4 · Edit plot scripts (minus-mean)

File: `talent/re_entry/army_basic_plots.py` — line 42:

```
COL_L00 = "pool_minus_mean_snr_fwd"
```

File: `talent/re_entry/army_hero_slide_plot.py` — line 44:

```
plot_var = "z_pool_minus_mean_snr_fwd"
```

File: `talent/re_entry/army_act2_probes.py` — line 55:

```
COL_L00_Z = "z_pool_minus_mean_snr_fwd"
```

Save all edited files.

Run 4 — Step 5 · Do not run the notebook

Skip Cells 0 through 11.

Run 4 — Step 6 · Check feather columns

```
python3 -c "import pandas as pd; df=pd.read_feather('big_dfs/df_pipeline_11_cox_analysis.feather');
print('z_pool_minus_mean_snr_fwd', 'YES' if 'z_pool_minus_mean_snr_fwd' in df.columns else 'MISSING')"
```

Expect: `YES`. If `MISSING`, the Run 1 notebook did not reach **Cell 10.5** — re-run Run 1 notebook, then repeat Run 1 archive and restore steps.

Run 4 — Step 7 · Edit manifest panel 9

Open `talent/re_entry/manifests/army_run1_3x3_manifest.json`.

Set panel 9 `"path"` to:

```
talent/re_entry/output/hero/ARMY_HERO_ew8_z_pool_minus_mean_snr_fwd_run1.png
```

Save the manifest.

Run 4 — Step 8 · Run plot scripts and mosaic

```
python talent/re_entry/army_basic_plots.py --all
python talent/re_entry/army_act2_probes.py --plot all_probes
python talent/re_entry/army_hero_slide_plot.py --plot-var z_pool_minus_mean_snr_fwd
python talent/re_entry/build_army_data_story.py
```

Run 4 — Step 9 · Verify Run 4

Panel 5 **H_sort** should **match Run 1** (same legacy grain in feather — only panels 3, 7, 8, 9 peer axes change).

Panel 9 porch may differ from Run 1 straight leave-one-out HERO.

Run 4 — Step 10 · Archive plots

```
bash scripts/backup_rename_suffix.sh _run4_legacy_minusmean --all-output
```

RUN 5 · Eval-thru pool grain · minus-mean plots · plot-only (no notebook)

What this run isolates: Minus-mean peer context on plots, using the **same feather** as Run 2 (eval-thru grain, keep zero-TB peers). **Do not re-run the notebook** unless the Run 2 archive is missing.

Run 5 — Step 1 · Open terminal on AWS

```
cd /path/to/Network_1P_shell
conda activate TALNET39
date
ls -lh big_dfs/df_pipeline_11_run2_thru.feather
```

If the archive file **does not exist**:

1. Re-run **Run 2** notebook (Run 2 Steps 2–4), then

2. Run Run 2 Step 9 feather save:

```
cp big_dfs/df_pipeline_11_cox_analysis.feather big_dfs/df_pipeline_11_run2_thru.feather
```

Run 5 — Step 2 · Restore Run 2 feather into the live slot

```
cp big_dfs/df_pipeline_11_run2_thru.feather big_dfs/df_pipeline_11_cox_analysis.feather
```

Direction: run2_thru → cox_analysis (restore, not save).

Run 5 — Step 3 · pipeline_config.py

No edit required for plot-only Run 5 — pool columns are already in the restored feather.
For consistent panel 5 labels only (optional, does not change feather):

```
POOL_GROUPING_MODE = 'active_at_eval_thru'  
POOL_EXCLUDE_PEER_TB_ZERO = False
```

Run 5 — Step 4 · Edit plot scripts (minus-mean)

File: talent/re_entry/army_basic_plots.py — line 42:

```
COL_L00 = "pool_minus_mean_snr_fwd"
```

File: talent/re_entry/army_hero_slide_plot.py — line 44:

```
plot_var = "z_pool_minus_mean_snr_fwd"
```

File: talent/re_entry/army_act2_probes.py — line 55:

```
COL_L00_Z = "z_pool_minus_mean_snr_fwd"
```

Save all edited files.

Run 5 — Step 5 · Do not run the notebook

Skip Cells 0 through 11.

Run 5 — Step 6 · Check feather columns

```
python3 -c "import pandas as pd; df=pd.read_feather('big_dfs/df_pipeline_11_cox_analysis.feather');  
print('z_pool_minus_mean_snr_fwd', 'YES' if 'z_pool_minus_mean_snr_fwd' in df.columns else 'MISSING')"
```

Expect: YES .

Run 5 — Step 7 · Edit manifest panel 9

Open talent/re_entry/manifests/army_run1_3x3_manifest.json .
Set panel 9 "path" to:

```
talent/re_entry/output/hero/ARMY_HERO_ew8_z_pool_minus_mean_snr_fwd_run1.png
```

Save the manifest.

Run 5 — Step 8 · Run plot scripts and mosaic

```
python talent/re_entry/army_basic_plots.py --all  
python talent/re_entry/army_act2_probes.py --plot all_probes  
python talent/re_entry/army_hero_slide_plot.py --plot-var z_pool_minus_mean_snr_fwd  
python talent/re_entry/build_army_data_story.py
```

Run 5 — Step 9 · Verify Run 5

Panel 5 H_sort should match Run 2 (same eval-thru grain in feather).
Compare panel 9 to Run 4 — minus-mean at eval-thru grain vs minus-mean at legacy grain.

Compare panel 9 to Run 2 — minus-mean vs straight leave-one-out at **eval-thru** grain.

Run 5 — Step 10 · Archive plots

```
bash scripts/backup_rename_suffix.sh _run5_thru_minusmean --all-output
```

Results table — fill in for Alex

Run	Pool grain	Exclude zero-TB peers?	Peer metric on plots	H_sort (panel 5)	HERO porch note (panel 9)
1 legacy	legacy	no	straight leave-one-out		
2 eval-thru	active_at_eval_thru		straight leave-one-out		
3 eval-thru no-zero	active_at_eval_thru	yes	straight leave-one-out		
4 legacy (archived feather) minus-mean	legacy	no	minus-mean		
5 eval-thru minus-mean	active_at_eval_thru (archived feather)		minus-mean		

Common mistakes

Symptom	Cause	Fix
HERO error: column not in feather ...run1.png	Passed filename to --plot-var	Use column name only, e.g. pool_tb_ratio_mean_snr_fwd
HERO error: z_pool_tb_ratio_mean_snr_fwd missing	Wrong column for Runs 1–3	Use pool_tb_ratio_mean_snr_fwd
Mosaic FileNotFoundError on panel 9	Manifest path ≠ HERO output filename	Edit manifest "path" to match exactly
Mosaic panels do not match standalone PNGs	Viewing backup subfolder PNGs	Open top-level talent/re_entry/output/... only
Mosaic build ran but looks old	Ran bash build_army_data_story.py	Use python talent/re_entry/build_army_data_story.py
Run 4 wrong grain	Forgot feather restore	cp run1_legacy.feather → cox_analysis.feather
Run 5 wrong grain	Missing or wrong archive	Save Run 2 feather in Run 2 Step 9; restore in Run 5 Step 2
Changed pipeline_config.py but feather unchanged	Did not re-run Cell 5–11	Re-run notebook after config change
Panel 1 shows N=39,517 but panel 2 shows n≈15,977	Different filters (cohort vs analysis sample)	Expected — not a compositor bug
Panel 2 zero spike after Run 3 exclude	Own talent zero, not peer exclusion	Expected — toggle drops peers with TB=0 only

Related documents

- Pool mode detail: talent/re_entry/rating_window_mode_explanation.md
- Straight leave-one-out mosaic notes: talent/re_entry/STRAIGHT_LOO_MOSAIC_AWS_GUIDE.md